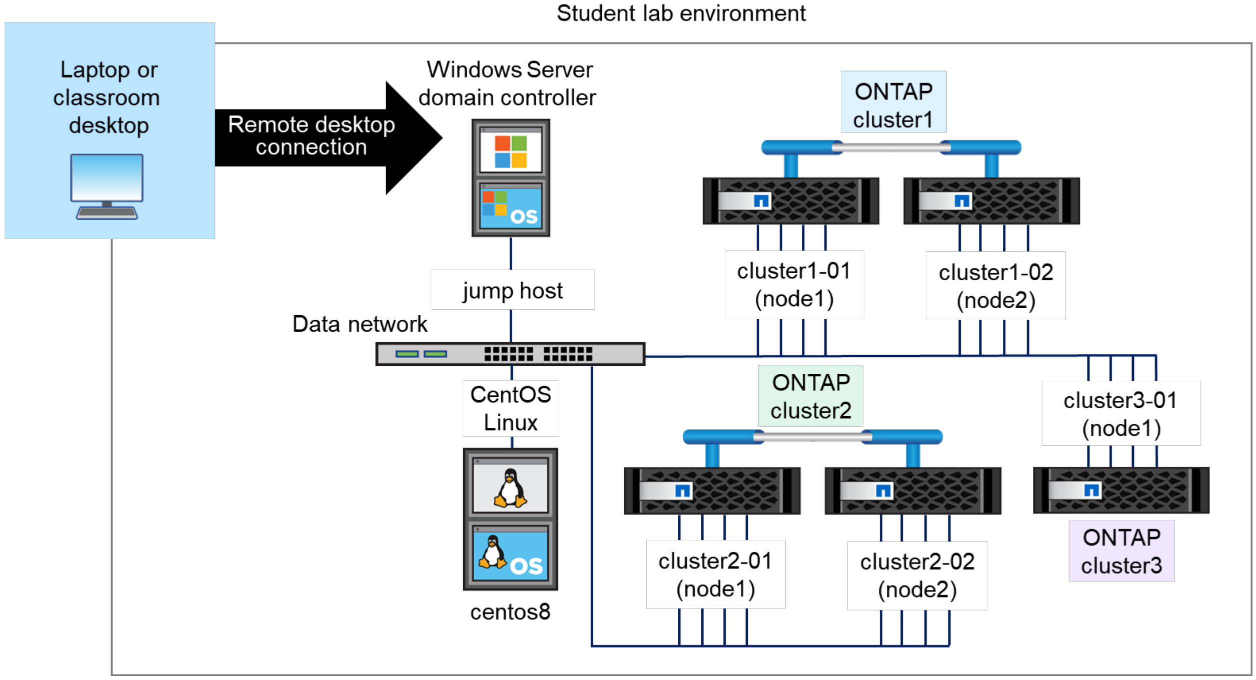
Module 3: SnapMirror Synchronous for Disaster Recovery

Exercise Equipment Diagram



System	Host Name	IP Addresses	User Name	Password
Windows Server 2022 Datacenter	JUMPHOST	192.168.0.5	DEMO\Administrator	Netapp1!
ONTAP cluster-management LIF (Cluster1)	Cluster1	192.168.0.101	admin (case-sensitive)	Netapp1!
node 1 (Cluster1)	cluster1-01-console	192.168.0.111	admin (case-sensitive)	Netapp1!
node 2 (Cluster1)	cluster1-02-console	192.168.0.112	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster2)	Cluster2	192.168.0 102	admin (case-sensitive)	Netapp1!
node 1 (Cluster2)	cluster2-01-console	192.168.0 113	admin (case-sensitive)	Netapp1!
node 2 (Cluster2)	cluster2-02-console	192.168.0.114	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster3)	Cluster3	192.168.0.103	admin (case-sensitive)	Netapp1!
node 1 (Cluster3)	cluster3-01-console	192.168.0.115	admin (case-sensitive)	Netapp1!
Linux Server	centos8	192.168.0.21	root	Netapp1!

Exercise 1: Configuring a SnapMirror Synchronous Relationship

In this exercise, you configure a SnapMirror synchronous (SM-S) relationship between two volumes.

Objectives

This exercise focuses on enabling you to do the following:

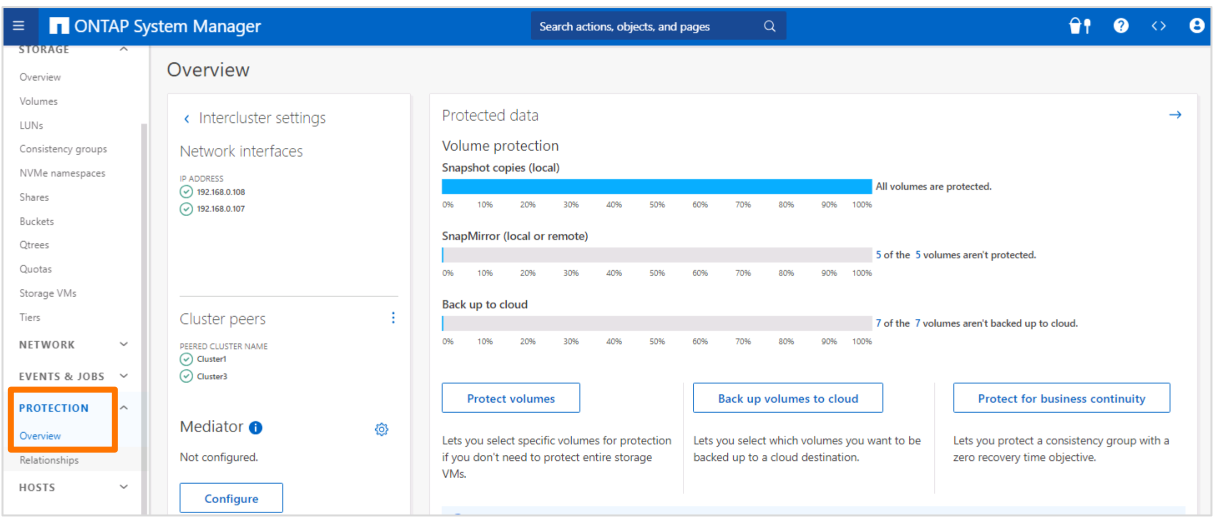
- Create an SM-S replication policy
- Create an SM-S relationship

Task 1: Create an SM-S Policy

In this task, you create an SM-S policy and relationship.

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1-1</td><td>

In the System Manager navigation pane for Cluster2, select **Protection > Overview**.



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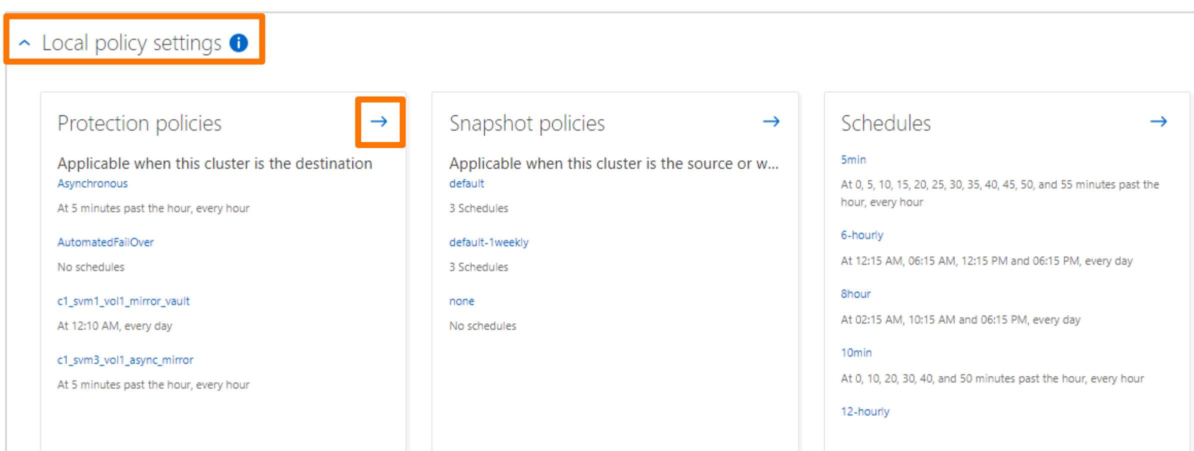
<tr><td>1-2</td><td>

In the Overview pane, expand **Local policy settings**, and then click the arrow in the upper-right corner of **Protection policies**.



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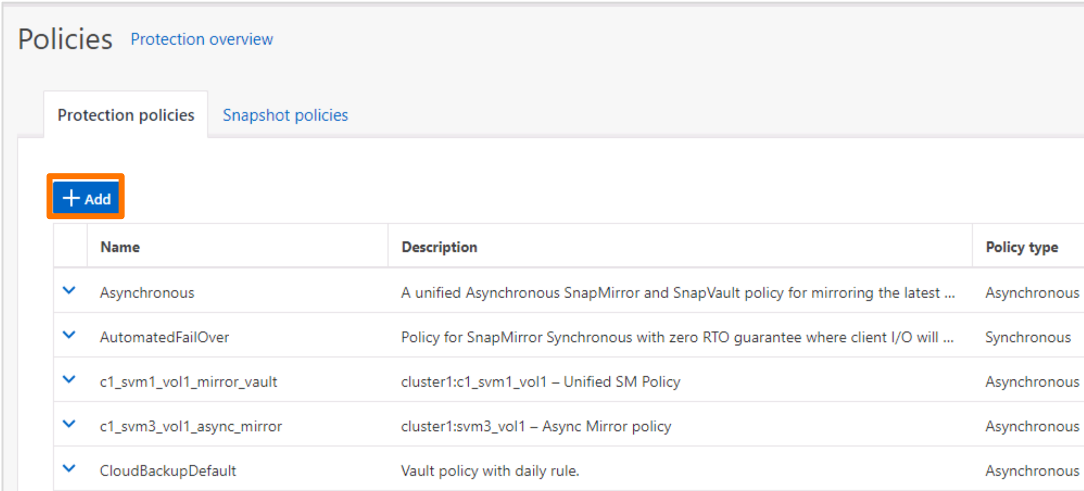
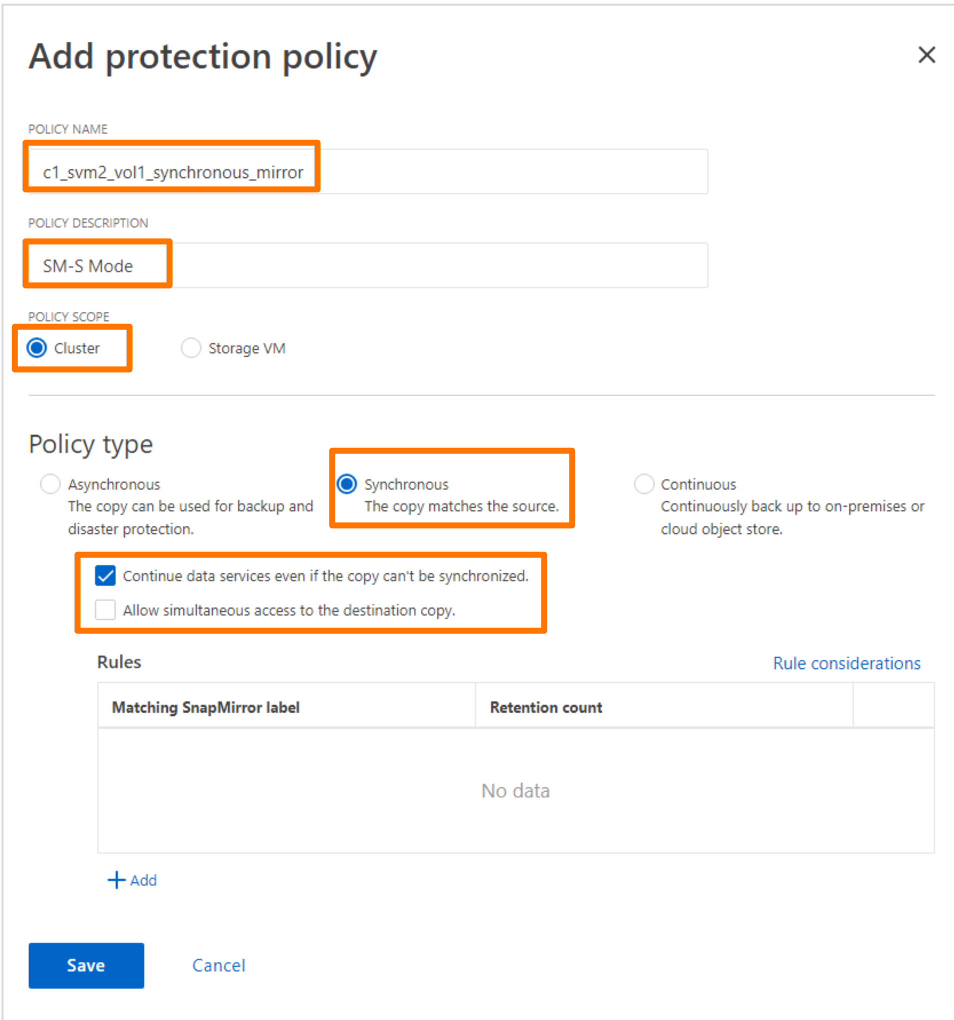
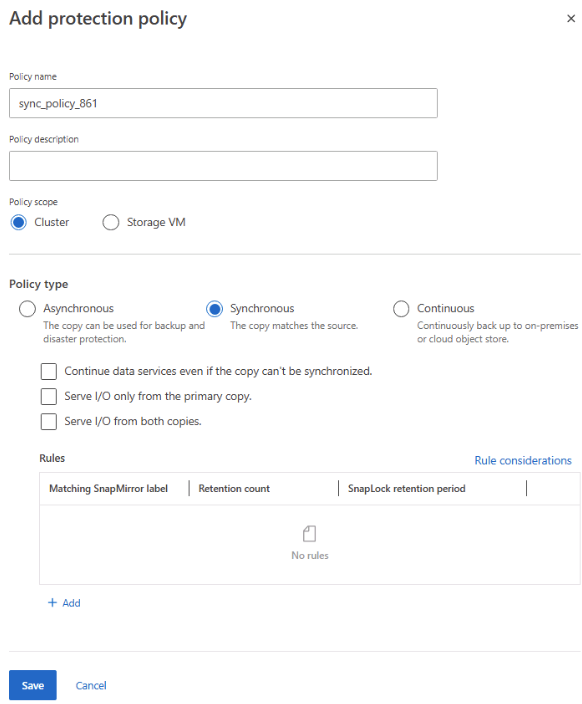
Step	Action
1-3	On the Protection policies tab, click **Add** to create a policy. 
1-4	In the following step, you select the Synchronous policy type before entering the policy name. If you select the Synchronous policy type after entering the policy name, the policy name changes to a default value. You then need to reenter the policy name.

1-5	In the Add protection policy dialog box, specify the following values:

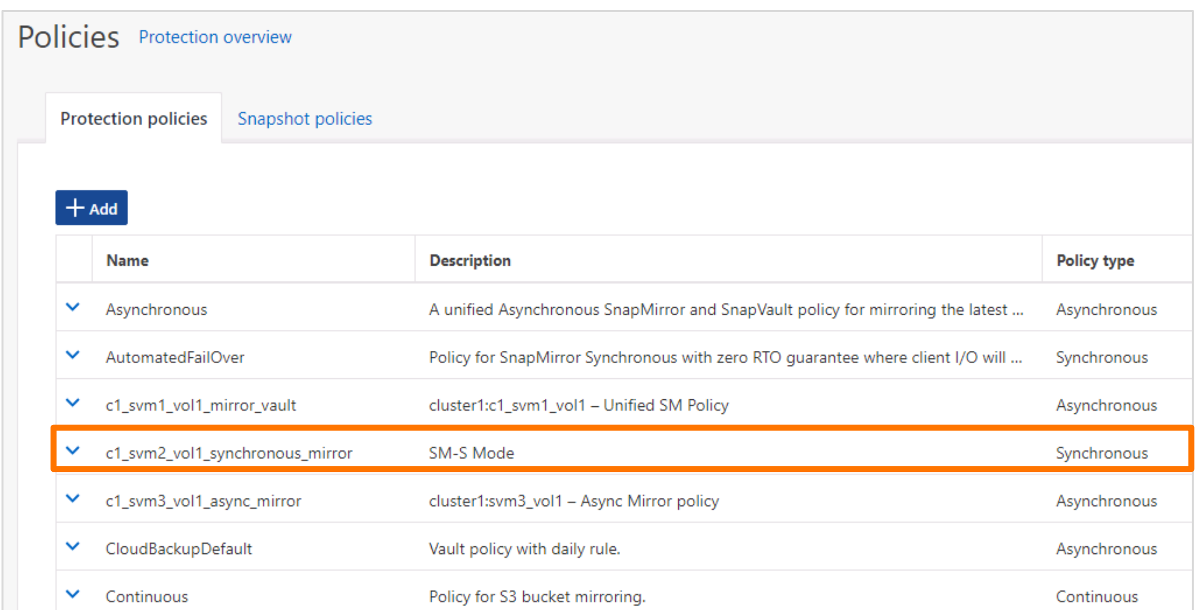
- Policy type: **Synchronous**
- Policy Name: **c1_svm2_vol1_synchronous_mirror**
- Policy Description: **SM-S Mode**
- Policy Scope: **Cluster**
- Continue data services even if the copy can't be synchronized: *Select checkbox*
- Allow simultaneous access to the destination copy: *Clear checkbox*



Step	Action

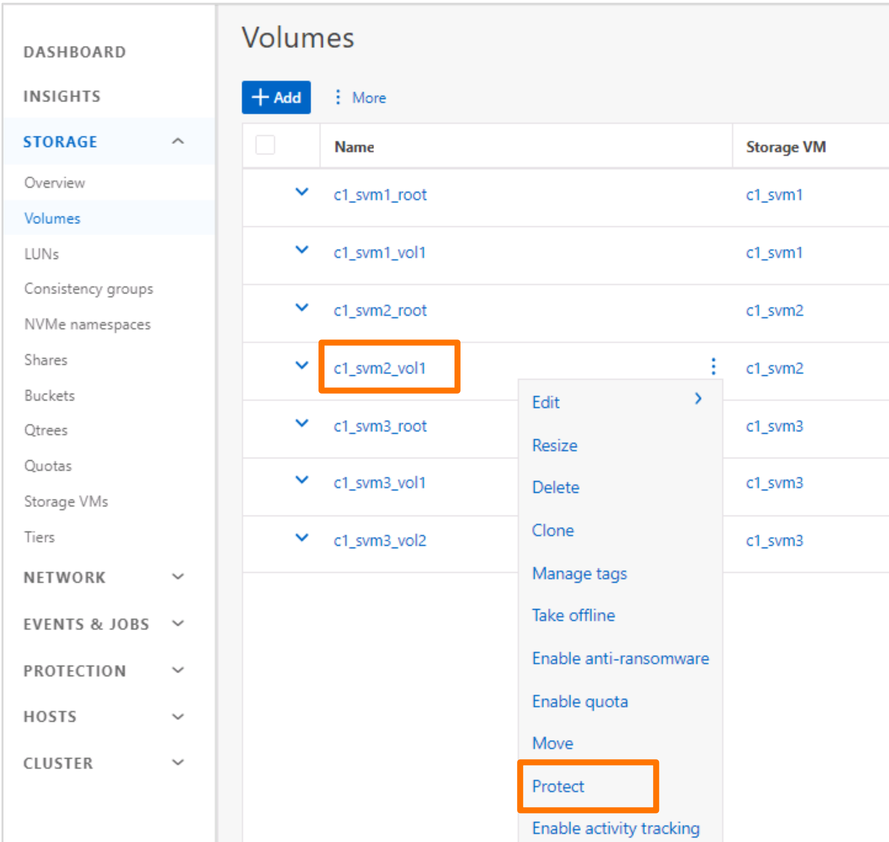


Step	Action
1-6	Click **Save**.
1-7	Synchronous policy is enabled by selecting Synchronous for the policy type and selecting the Continue data services even if the copy can't be synchronized checkbox.
1-8	On the Protection policies tab, verify that the policy was created successfully.

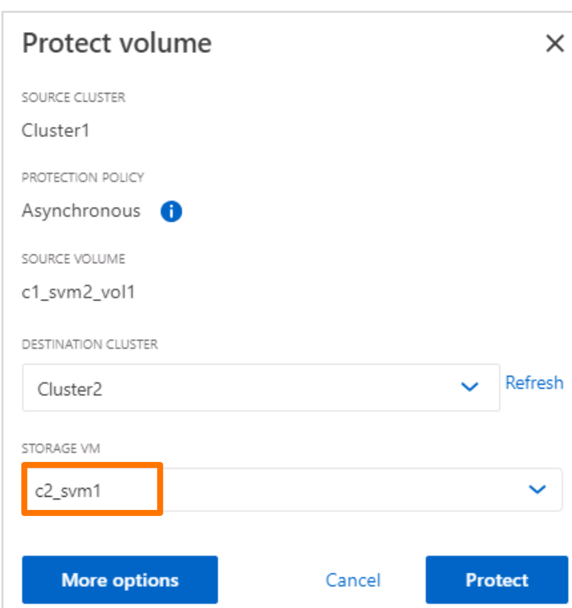


Task 2: Create an SM-S Relationship

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>2-1</td><td>In the System Manager navigation pane for Cluster1, select Storage > Volumes.</td></tr>
<tr><td>2-2</td><td>Click the three vertical dots to the right of c1_svm2_vol1 and then select Protect.



</td></tr>
<tr><td>2-3</td><td>In the Protect volume dialog box, select c2_svm1 for the destination SVM.



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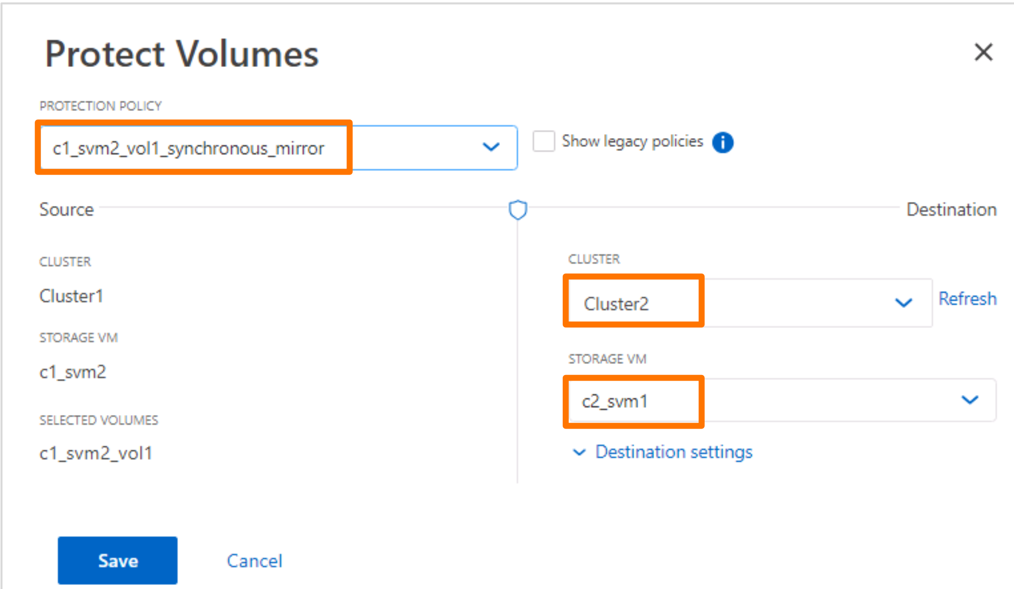
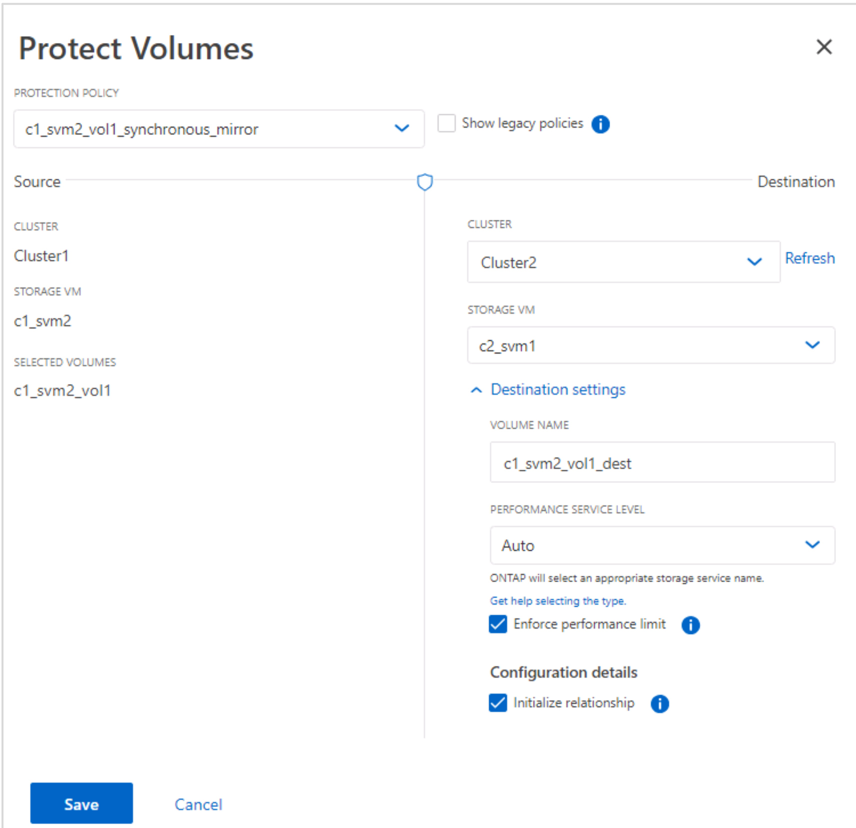
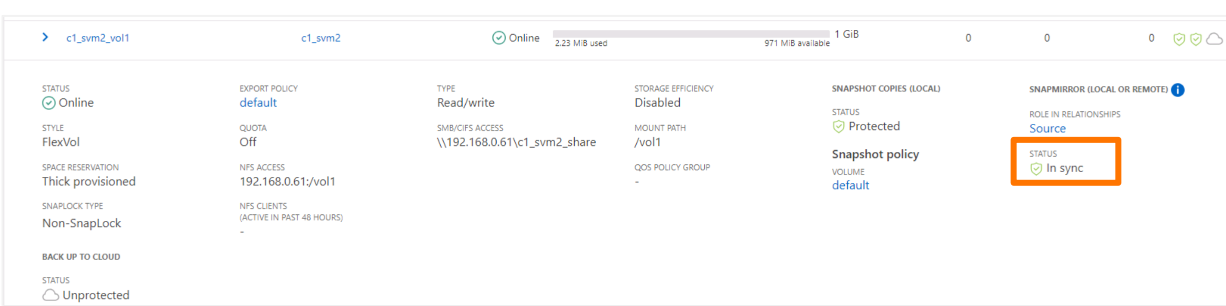
Step	Action
2-4	Click **More options**.
2-5	In the Protect Volumes pane, specify the following values: ▪ Protection Policy: **c1_svm2_vol1_synchronous_mirror** ▪ Destination > Cluster: **Cluster2** ▪ Destination > Storage VM: **c2_svm1** 

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>2-6</td><td>Expand Destination settings and then review the values that are listed without making any changes.

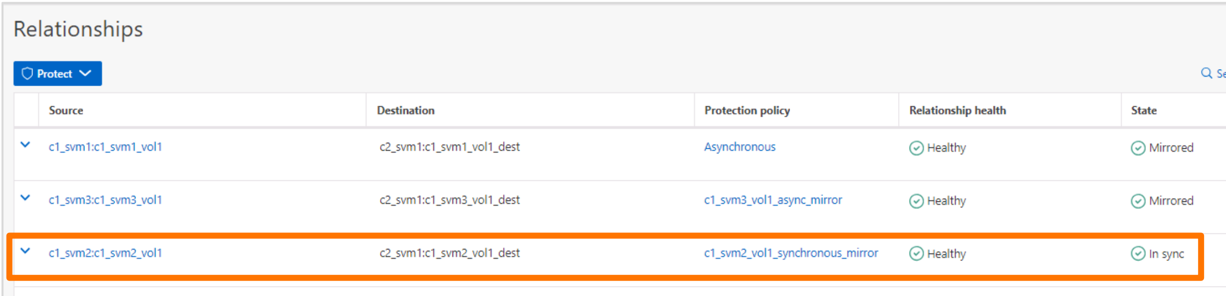


</td></tr>
<tr><td>2-7</td><td>Click Save.</td></tr>
<tr><td>2-8</td><td>i Cluster1 and Cluster2 are currently in a peer relationship. On Cluster1, c1_svm2 is not in a peer relationship with c2_svm1 on Cluster2. However, after you click Save in the previous step, System Manager automatically peers the two SVMs and then creates the SnapMirror relationship.</td></tr>
<tr><td>2-9</td><td>i The Adding the relationship status bar appears for a few seconds. Optionally, click Run in Background.</td></tr>
<tr><td>2-10</td><td>In the Volumes pane, click the down arrow next to c1_svm2_vol1 to view the details and verify that the protection relationship was created successfully, and see that the status is In sync.

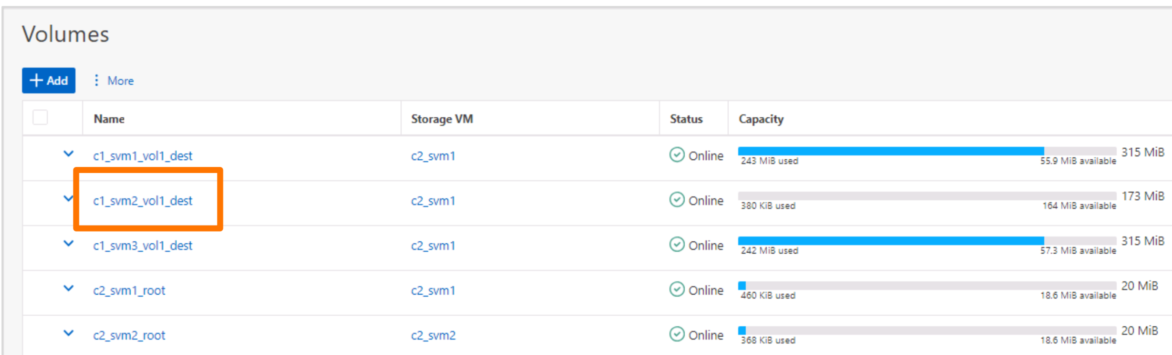


</td></tr>
</table>

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>2-11</td><td>In the System Manager navigation pane for Cluster2, select Protection > Relationships.</td></tr>
<tr><td>2-12</td><td>In the Relationships pane, verify that the SM-S relationship is listed and that the Relationship health is Healthy.



</td></tr>
<tr><td>2-13</td><td>In the navigation pane, select Storage > Volumes.</td></tr>
<tr><td>2-14</td><td>In the Volumes pane, verify that the destination volume for the SM-S relationship is listed.



</td></tr>
<tr><td>2-15</td><td>i Optionally, you can map a drive to the source volume and copy File_240M from C:\CourseFiles to the mapped drive. After the data transfer, you can review the storage capacity that is used by the destination volume.</td></tr>
</table>

End of exercise